Theory and Epistemology of Digital Humanities

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Today I would like to talk about the theory and epistemology of digital humanities. In very general terms, our starting point is the observation that so far, digital humanities has largely contented itself with borrowing methods from other fields and has developed little methodology of its own. This borrowing of methods is often very pragmatic—some may say: opportunistic—and rarely accompanied by theoretical reflection. The consequence is that there is effectively no epistemology of digital humanities; in other words: there is little reflection on how knowledge is produced in digital humanities and whether the production of knowledge in DH differs from that of the traditional humanities—and if so, in what respect.

1 Digital Humanities, or: Close Encounters of the Third Kind

This is particularly striking because "digital humanists"—especially from the Anglo-American school—do not seem to tire stressing the alienness of computational methods to the humanities and their immense epistemological impact. For example, Berry and Fagerjord (2017) note:

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> [S]ome of the key techniques that are deployed in the digital humanities [...] importantly raise questions for epistemology in relation to knowledge production and analysis.
> —Berry and Fagerjord (2017, 105)

In fact, according to Burdick et al. (2012):

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> The foundational layer, computation, relies on principles that are, on the surface, at odds with humanistic methods.
> —Burdick et al. (2012, 16)

And Galloway (2014) warns:

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> Few things will cripple the humanities more than the uncritical "adoption of tools" or the continued encroachment of positivistic research methods borrowed from cognitive science, neuroscience, computer science, or elsewhere.
> —Galloway (2014, 128)

Consequently, Drucker (2011) exhorts:

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> [D]igital humanities can no longer afford to take its tools and methods from disciplines whose fundamental epistemological assumptions are at odds with humanistic method.
> —Drucker (2011, para. 6)

What we find here, *nota bene* among proponents of digital humanities, is a fundamental unease with the "tools and methods" that presumably characterize digital humanities. Computational methods are essentially regarded as intrusion of a *useful but lesser mode of inquiry* into the humanities.

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> While the research questions may be similar to those in the "analogue humanities," the constitution of what counts as an interesting answer may have changed so profoundly that it no longer is compatible with the traditional epistemology of the humanist disciplines.
> —Berry and Fagerjord (2017, 21)

And Dobson (2019) affirms that these conflicts between computation and "the traditional epistemology of the humanist disciplines" are not only still not resolved, but in fact "fundamentally unresolvable."

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> [...] not only are these conflicts still not resolved, but they remain fundamentally unresolvable.
> —Dobson (2019, 6)

The issues raised here seem pretty serious to me, but, and I find this quite remarkable, none of the authors actually feel the need to tell us what "the traditional epistemology of the humanist disciplines" (Berry and Fagerjord 2017, 21) is, what "the methods of the sciences" (Dobson 2019, 6) refers to in this context, or how any cooperation may be possible with "disciplines whose epistemological foundations and fundamental values are at odds with, or even hostile to, the humanities" (Drucker 2012, 85).

In fact, there is very little research on the epistemological foundations of digital humanities and on how they differ from the "traditional" humanities; in other words, research that tries to answer the question: what is the impact of computational methods on the production of knowledge in the humanities?

There seem two main obstacles to even start thinking about these questions. First of all, as Abraham Moles ([1990] 1995) remarked, there actually is not much in terms of an epistemology of the humanities.

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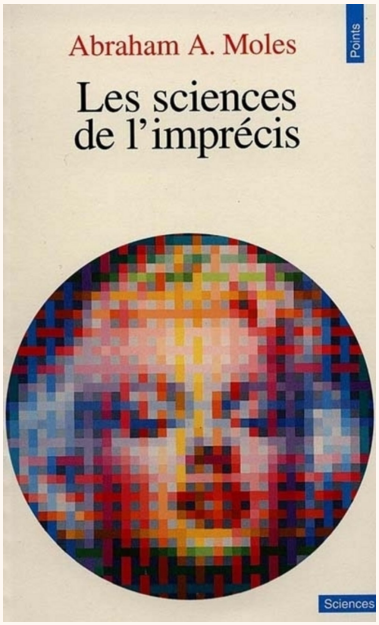
> [L]'épistémologie des sciences humaines n'existe pas encore.
> —Moles ([1990] 1995, 46)



Of course, there is, for example Hans-Georg Gadamer, and many humanities scholars would argue that "the hermeneutic method," i.e., interpretation, is how the humanities produce knowledge. This is, however, a misunderstanding. In *Wahrheit und Methode* ([1986] 1993) Gadamer rejects Dilthey's conception of the *Geisteswissenschaften* on the model of the *Naturwissenschaften*, as well as his idea of a kind of methodological parallelism (Wright 1976, 415) between them. For Gadamer, interpretation not a method in the scientific sense, something that could be taught. Instead, it's all about *bildung* and tradition:

> The modern concept of science and the associated concept of method are insufficient. What makes the human sciences into sciences can be understood more easily from the tradition of the concept of Bildung than from the modern idea of scientific method. It is to the *humanistic tradition* that we must turn. In its resistance to the claims of modern science it gains a new significance.
> —Gadamer (1989, 16)

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> Der moderne Wissenschaftsbegriff und der ihm zugeordnete Methodenbegriff können nicht ausreichen. Was die Geisteswissenschaften zu Wissenschaften macht, läßt sich eher aus der Tradition des Bildungsbegriffes verstehen als aus der Methodenidee der modernen Wissenschaft. Es ist die *humanistische Tradition*, auf die wir zurückverwiesen werden. Sie gewinnt im Widerstand gegen die Ansprüche der modernen Wissenschaft eine neue Bedeutung.
> —Gadamer ([1960] 1990, 23)

In fact, he argues that unlike the natural sciences, the humanities don't have fixed research goals and that the notion of progress doesn't make sense for them either, because, according to Gadamer, research results in the humanities become practically never obsolete:

> A modern reader can easily make allowances for the fact that, a hundred years ago, less knowledge was available to a historian, and he therefore made judgments that were incorrect in some details. On the whole, he would still rather read Droysen or Mommsen than the latest account of the subject from the pen of a historian living today.
> —Gadamer (1989, 285)

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> Der heutige Leser weiß offenbar mit Leichtigkeit davon zu abstrahieren, daß ein Historiker vor hundert Jahren über einen geringeren Wissensbestand verfügte und deshalb in manchen Einzelheiten verkehrte Urteile fällte. Aufs Ganze gesehen wird er immer noch Droysen oder Mommsen lieber lesen als die neueste Darstellung der betreffenden Gebiete aus der Feder eines gegenwärtig lebenden Historikers.
> —Gadamer ([1960] 1990, 287–88)

Thus, I believe that Gadamer isn't really a good foundation for an epistemology of the humanities, or at least none that is useful to us.

Second, a large share of the people in digital humanities persistently refuse to define what this field is about. This is of course not news, but just to give you a very recent example, here is an excerpt of an interview Valérie Schafer from the University of Luxembourg gave in March 2022:[1]

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> **Rabea Rittgerodt:** How would you define Digital Humanities?
> **Valérie Schafer:** That's very easy—absolutely no controversy and discussion about it! (laughs) No, of course I cannot give you a unique definition because there are a lot of people trying to define digital humanities and it's quite complex. I would prefer not to give a precise definition, but rather to say that for me it's an interdisciplinary field, a transversal one. It's a community—or several communities. […]
> To summarize, it's a multi- and interdisciplinary field at the intersection of humanities and computer science, which is based on a shared passion for research, creativity, new ways of working, and values. […]
> (Digital Humanities – A Science unto Itself? A Conversation with Valérie Schafer)

So, digital humanities is all about community, shared passion, creativity, and values? This sounds like just a bunch of people hanging out together and having fun.

If you ask me, this view is a big problem in many ways when we are talking about research rather than nightlife, and it is most definitely a problem with respect to the epistemology: if you cannot say what you are doing, and why you are doing it, it will be hard to say how you are producing new knowledge. It will also be hard to justify why taxpayers should fund you.

2 Defining Digital Humanities

So, in order to discuss the epistemology of DH, we need to say what we mean by DH. I won't go into details here, but we need a definition if we want to move on. So, I will just give you my definition (Piotrowski 2018; Piotrowski and Xanthos 2020), and I hope it will become clear why I propose this definition:

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Digital Humanities

1. The **applied digital humanities** are concerned with the construction of formal—computational—models of the phenomena studied by their "mother disciplines," as well as the methodology of their construction.
2. The **theoretical digital humanities** study the general properties of formal—computational—models in the humanities at a higher level of abstraction.

As you can see, I distinguish between applied and theoretical digital humanities. It is crucial to make this distinction, because their research objects and research objectives are very different.

[1] Source: Digital Humanities – A Science unto Itself? A Conversation with Valérie Schafer, March 10, 2022.

So, the theoretical digital humanities create and study the *meta*models whose concrete application to research questions in the disciplines of the humanities is the object of the applied digital humanities.

3 Models

Clearly the notion of *model* is central to my understanding of digital humanities. Let me briefly explain what I mean by "model." Even in the natural sciences, the explicit notion of *model* is surprisingly recent, and only began to be studied seriously in the 1950s. One of the first "modern" attempts at a *general* definition of models is the following by the Belgian philosopher Leo Apostel:

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> [...] *any subject using a system A that is neither directly nor indirectly interacting with a system B to obtain information about the system B*, is using A as a model for B.
> —Apostel (1961, 36)

I think this is still an excellent, concise definition. And it does not just apply to science, but holds generally. This cartoon, by Irina Blok, perfectly illustrates the fact that what we have in our heads is not "reality," but merely a *model* of it.

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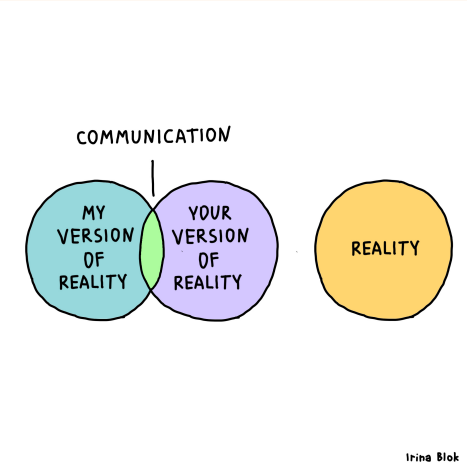

The purpose of science, though, is to build *intersubjective models* that can be communicated and tested against "reality." As Gilles-Gaston Granger ([1960] 1967) notes, the goal of any science (natural or other) is to build *coherent and effective models of the phenomena they study*.

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> Quant à l'intentionnalité scientifique, à la visée, nous l'avons déjà définie comme construction de *modèles cohérents et efficaces du phénomène*.
> —Granger ([1960] 1967, 215)

As Jean-Pierre Dupuy wrote, there is much debate in history and philosophy of science about the nature and foundations of scientific knowledge, but "the fact that science as an *activity* consists essentially in constructing objects in the form of models is incontestably true, even if it is not as well known among nonscientists as it should be" (Dupuy [2000] 2009, 29).

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> On dispute beaucoup en philosophie et histoire des sciences de la nature et des fondements de la connaissance scientifique. Que la science comme *activité* consiste essentiellement à se construire des objets sous la forme de modèles est en revanche une vérité incontestable, bien que trop peu connue des non-scientifiques.
> —Dupuy ([1999] 2005, 18)

What we have then is a setting that I have recently started calling the *modeling triad*.

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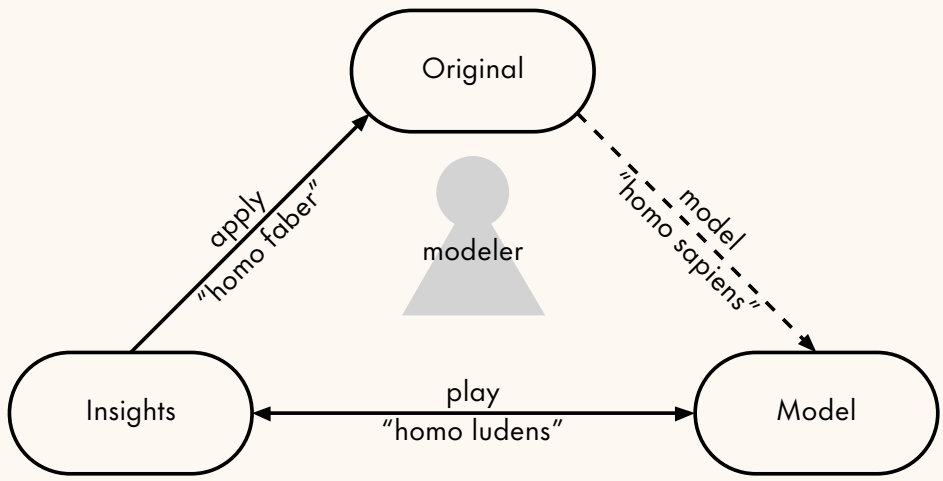

First, there is the original, the phenomenon that we want to study and on which we aim to gain insights. The original is not directly and completely accessible to us—something which is obviously true for historical phenomena, and which is precisely why we *model*.

The second step is to *play* with the model. Playing can mean, for example, to change the value of a parameter or to add or remove some component, depending on the nature of the model. In this context, *playing* should be understood as a technical term for manipulating a model (as used, e.g., by Klaus 1966, 93ff), in contrast to *experimenting*, which refers to *strong interaction* (Moles [1990] 1995, 100), i.e., the manipulation of the original. Our hope is that playing with the model will yield new insights on the original.

Third, we *apply* our insights gained from our interaction with the model to the original. "Apply" clearly has to be understood in a very wide sense, as the nature of the application will be very different in different domains. In the natural sciences, a prediction made on the basis of a model can be compared to the actual outcome; in engineering, design decisions may be taken, but none of this is generally possible in the humanities, so the *viability* of the model will have to be evaluated in different ways. I say "viability," because, and let me stress this again, a model cannot be "compared" to the original. This would require an objective observer, and as Heinz von Foerster put it, objectivity is a subject's delusion that observing can be done without him.

The terms *homo sapiens*, *homo ludens*, and *homo faber* are references to the *epistemological control loop* (erkenntnistheoretischer Regelkreis) by Georg Klaus (1966, 93).

4 Models in the Humanities

The big question is now: what are models in the humanities, and computational models in particular?

Some disciplines in the humanities have a long tradition of using explicit models (sometimes even semi-formal or formal models) under various names, for example grammars in linguistics. Other disciplines are openly hostile to this notion. With respect to historians, for example, Gordon Leff (1972) observes:

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> Historians as a profession are not given to constructing or employing models in any formal or explicit sense; where they do, it is mainly in areas bordering on other disciplines, especially economics and social studies. Most historians, if asked, would probably deny that models had anything to do with their subject.
> —Leff (1972, 148)

In fact, he points out, historians could hardly put pen to paper without having implicit models of what they were studying. Nevertheless, most research on models in philosophy of science has focused on models in the natural sciences.

In digital humanities, the importance of modeling is frequently acknowledged, but—and this is probably due to the fact that modeling is widely considered foreign to the humanities—typically as a specific, clearly delimited activity, which is needed for technical reasons. For example:

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> The use of models and modelling also has a long tradition in the humanities. [...] Nevertheless it is fair to say that the explicit use of the word "modelling" in humanities research has increased significantly with the introduction of Digital Humanities (hereafter DH), where modelling is considered one of the core research practices [...]. The high reliance on modelling in this discipline is due to the fact that explicit models are extensively required in DH in order to operationalise research questions.
> —Ciula et al. (2018, 346)

While I agree on many of these points, I would argue that the understanding of modeling as "one of the core research practices" is too limited and potentially misleading. As Joshua Epstein (2008) has remarked, the choice "is not whether to build models; it's whether to build explicit ones."

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> The choice, then, is not whether to build models; it's whether to build explicit ones. In explicit models, assumptions are laid out in detail, so we can study exactly what they entail. On these assumptions, this sort of thing happens. When you alter the assumptions that is what happens. By writing explicit models, you let others replicate your results.
> —Epstein (2008)

In actual research practice in DH, modeling is explicit and formal, as required by computational methods, but often lacking reflection. This becomes especially clear when we consider corpora.

5 Corpora

When we look for epistemological differences between "traditional" and digital humanities, *machine-readable corpora*—stand out. That they are machine-readable is today usually implied, so people just talk about corpora, and I will use the term in this sense as well, which has been likely inspired by its usage in computational linguistics. Of course, scholars have always created and studied collections of documents and artifacts; in fact, this could be said to be a significant aspect of scholarship.

What is clearly new, however, is that corpora—machine-readable corpora—are, well, machine-readable. This has obvious practical advantages: they are easier to store, and they can be processed automatically. What tends to be overlooked, though, is the epistemological impact. After all, it is not some menial task that is being automated: the main results of many DH projects is based on the automatic analysis of some corpus. The availability of a machine-readable corpus is the precondition for the use of computational methods, at least as they are commonly understood, i.e., quantitative methods.

While there is much discussion of methods and their appropriateness—and many common definitions as well as a large part of the criticism of DH are related to these methods—there is surprisingly little discussion of corpora in DH. In a typical DH paper (or project proposal), just a few words are said about the corpus that was used, and most of it tends to concern its size and composition (n items of class X, m items of class Y, and so on) and the technical aspects of its construction (e.g., how it was scraped), if the authors did not use an existing corpus.

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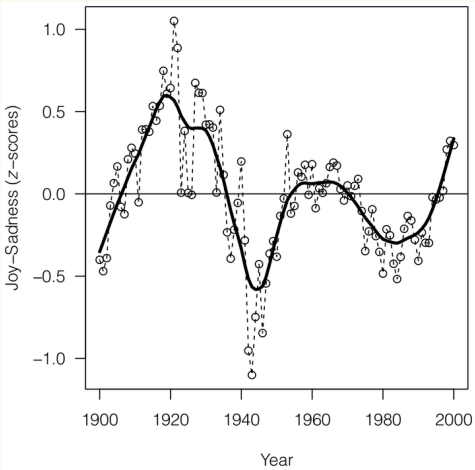
According to Acerbi et al. (2013), moods have followed broad historical trends, including a "sad" peak corresponding to World War II, and two "happy" peaks, one in the 1920s and one in the 1960s. There is also a "sad" period starting in the 1970s, with an increase in "happiness" in the later years of the data set.

▸ World War I does not seem to register any particular change in mood words.



The methods used and the results achieved (and their interpretation and visualization) are typically discussed extensively, though.

I would like to just highlight two epistemological problems here, which *should* be obvious, as they also apply to statistics in general, but apparently they are not.

The first one is that all these seemingly objective quantitative measures (n items of class X, m items of class Y, and so on) are crucially dependent on the preceding *qualitative* analysis that established the classes: the difference between an apple and an orange is not a quantitative, but a *qualitative* difference. What is more, it is in no way the only possible classification: classify apples, oranges, and pears as "fruit," or by color, size, or shape, or for that matter, acidity, and you may get completely different numbers.

So this all depends on how we analyze our reality—in the literal sense of the Ancient Greek ἀνάλυσις, "breakup," and there is more than one way to do it. Granger ([1960] 1967) calls this *le découpage des phénomènes*. As the English translation (Granger 1983) leaves the term *découpage* untranslated, I will do so as well, but in my opinion, there is really nothing inherently untranslatable about it.

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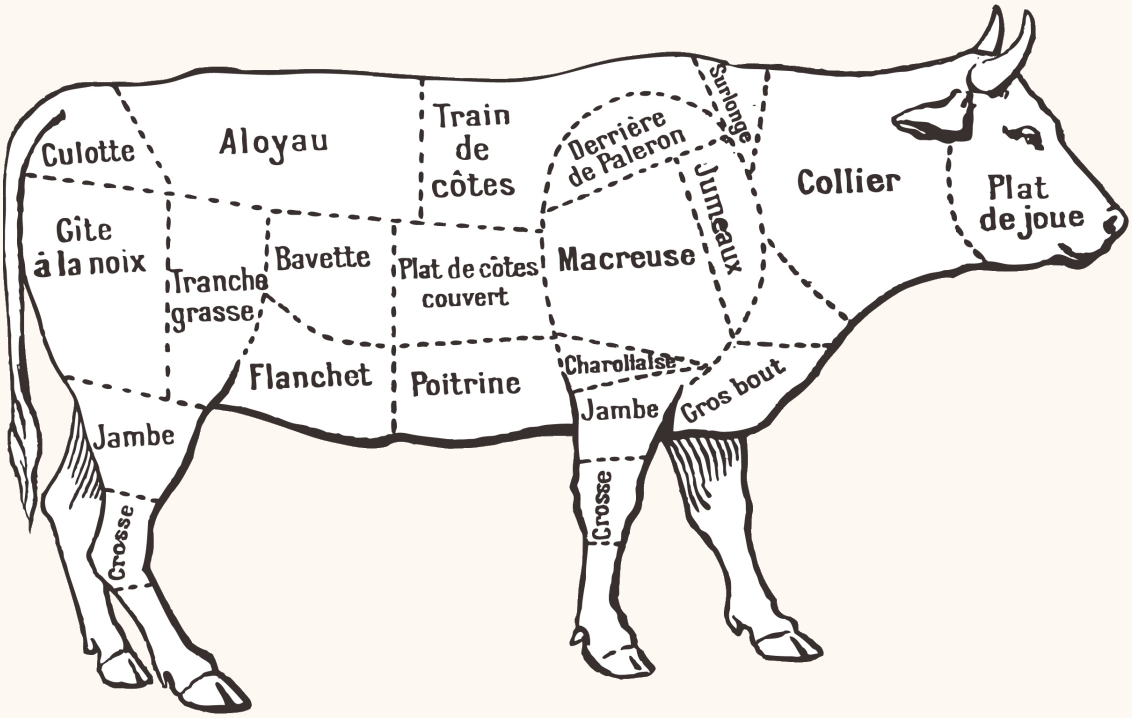

In any case, Granger notes that the *découpage* of human facts—i.e., those studied by the humanities—presents a special difficulty:

> Here the phenomena have an immediate *sense*, which means that they spontaneously take part in a universe of valorized and directed actions, either in the consciousness of an individual, or in the organization and functioning of a collectivity which is given as a whole, even when the relations of this whole escape us. This sense is transmitted by language for the speaking subject of each social group, and it is this that constitutes, for our consciousnesses as agents, the very essence of the given human fact.
> —Granger (1983, 49–50)

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> C'est que le découpage des faits humains présente une difficulté spécifique. Les phénomènes ont ici un *sens* immédiat, ce qui veut dire qu'ils font spontanément partie d'un univers d'actions valorisées et orientées, soit dans la conscience d'un individu, soit dans l'organisation et le fonctionnement d'une collectivité qui se donne comme un tout, alors même que les liaisons de ce tout nous échappent. Ce sens est véhiculé par le langage pour le sujet parlant de chaque groupe social, et c'est lui qui constitue pour nos consciences d'acteurs l'essence même du fait humain donné.
> —Granger ([1960] 1967, 64)

As a naive instrument of this *découpage* of facts, Granger argues, language naturally transmits what he calls *myths*, as well as ideologies, i.e., more or less explicit justificatory interpretations of situations actually realized by social organization and practice at a given moment of history. One of the first challenges is thus to overcome this overdetermination of notions in order to reach scientific concepts.

We can then think about what Moles ([1990] 1995, 110) considers the first step in the transition from the vague towards the exact: measurement, as a system of comparison between the subjective human impression and external material data, as in the example of a scale of roundness (as a function of the radius of curvature of the various asperities [rough edges] and angles of the particle considered)—in a way establishing an intersubjective scale between a "rock" and a "pebble."

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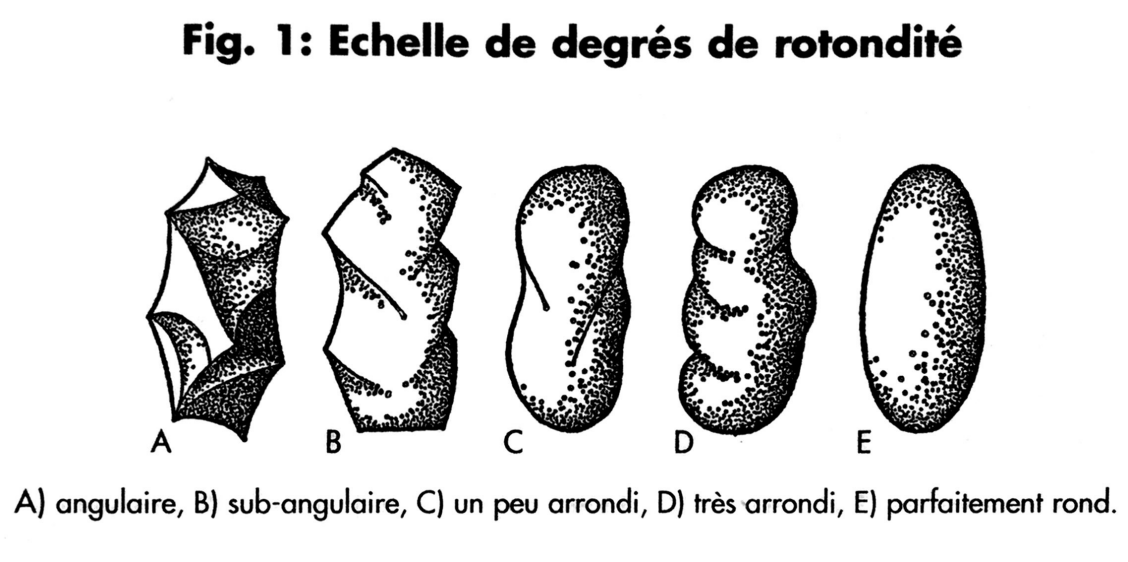

However, in general the humanities still have a long way to go in this respect. Granger ([1960] 1967, 66) notes that "the constitutive problem of the humanities can be described as the transmutation of lived meanings into a universe of objective meanings. Hence the fundamental importance, beyond the trials and tribulations, of new disciplines like information theory and cybernetics."

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> Le problème constitutif des sciences de l'homme peut être dès lors décrit comme transmutation des significations vécues en un univers de significations objectives. De là vient l'importance fondamentale, par-delà les tâtonnements et les insuffisances, de disciplines nouvelles comme la théorie de l'information et la cybernétique.
> —Granger ([1960] 1967, 66)

The second issue is, again, a general one, but which is particularly tricky in our context. In statistical terms, a corpus can be considered a *sample* of some *population* that cannot be analyzed in its entirety, say, all French-language novels published in the 19th century. The core idea of statistics is that by analyzing the sample, we can obtain results that can be generalized to tell us something about the whole population.

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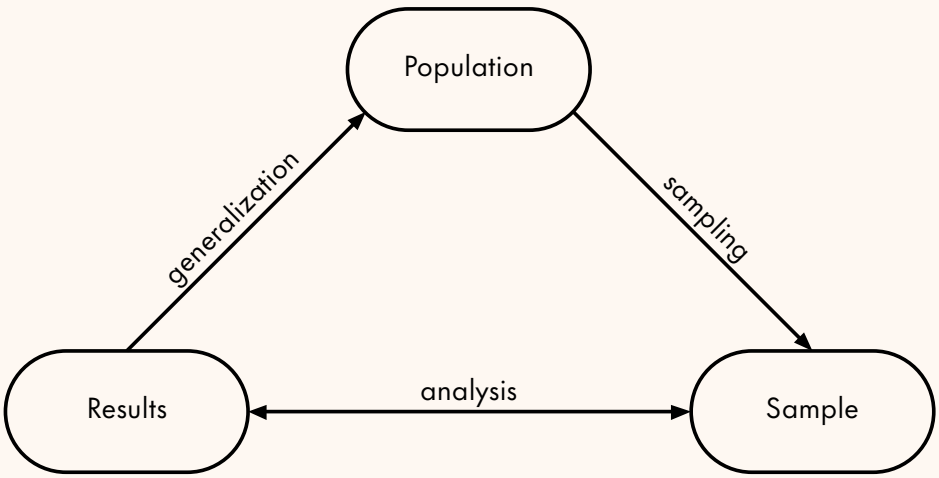

As a matter of fact, this is merely a special case of the modeling triad I showed you earlier. Thus, and as I have argued before (Piotrowski 2019), a corpus should be considered a model in the sense of Apostel (1961). Creating a corpus thus means constructing a model, and modelers consequently have to answer questions such as: What is the original? In what respects is the model a reduction of it? And for whom and for what purpose am I creating the model?

These are not new questions: every time historians select sources, they construct models, even before any detailed analysis. However, machine-readable corpora are not only potentially much larger than any material collection of sources—which is already not inconsequential—but also have important epistemological consequences. The larger and the more "complete" a corpus is, the greater the danger to succumb to an "implicit essentialism" (Mothon 2010, 19) and to mistake the model for the original, a fallacy that can frequently be observed in the field of cultoromics (Michel et al. 2011), when arguments are being made on the basis of the Google Books Ngram Corpus.

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> A map is *not* the territory it represents, but, if correct, it has a *similar structure* to the territory, which accounts for its usefulness.
> —Korzybski (1933, 58)

The same then goes for any analysis of a corpus: if the corpus is "true," so must be the results of the analysis; if there is no evidence of something in the corpus, it did not exist. This allure is even greater when the analysis is done automatically and in particular using opaque quantitative methods: as the computational analysis is assumed to be completely objective, there seems to be no reason to question the results—they merely need to be interpreted.

However, an analysis of a corpus will *always* yield results; the crucial question is whether these can tell us anything about the original phenomenon it aims to model.

The typical case of a corpus-based digital humanities project thus resembles an *experimental system* (Rheinberger 2001)—an appearance supported by the laboratory metaphor typical for the DH. However, this appearance is deceptive, as ultimately research objects, research questions, and research methods differ substantially from the natural sciences.

The crucial point is that corpora are not naturally occurring but intentionally constructed. A corpus is *already* a model and thus not epistemologically neutral. Here we build on Gaston Bachelard's notion of *phénoménotechnique* (Bachelard [1934] 1968).

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> La véritable phénoménologie scientifique est donc bien essentiellement une phénoménotechnique. […] Elle s'instruit par ce qu'elle construit.
> —Bachelard ([1934] 2020, 35)

Bachelard originally developed this notion, which treats scientific instruments as "materialized theories," as a way to study the epistemology of modern physics. As we have said before, the humanities have also always constructed the objects of their studies (to a large extent not directly observable, but only through artifacts, in particular texts) through, for example, the categorization and selection of sources and the (hermeneutic) postulation and affirmation of phenomena.

However, only the praxis has been codified to some extent as "best practices," such as source criticism. Theories—or perhaps better: models and metamodels, as the term "theory" has a somewhat different meaning in the humanities than in the sciences—are not formalized and are only suggested by the (natural language) narrative. What history (and the humanities in general) traditionally do not have is something that corresponds to the scientific instrument.

This changes with digitalization and "datafication": phenomena are now constructed and modeled through data and code, and (like in the sciences), the computational model takes on the role of the instrument and "sits in the center of the epistemic ensemble" (Rheinberger 2005, 320). Corpora are then, methodologically speaking, phenomenotechnical devices and form the basis and influence how we build, understand, and research higher-level concepts.

Thus, a corpus *produces* the phenomenon to be studied. As a model, it has Stachowiak's three characteristics of models (↗ Stachowiak 1973, 131–33):

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Die drei Hauptmerkmale des allgemeinen Modellbegriffs

1. **Abbildungsmerkmal:** Modelle sind stets Modelle *von etwas*, nämlich Abbildungen, Repräsentationen natürlicher oder künstlicher Originale, die selbst wieder Modelle sein können.
2. **Verkürzungsmerkmal:** Modelle erfassen im allgemeinen nicht alle Attribute des durch sie repräsentierten Originals, sondern nur solche, die den jeweiligen Modellerschaffern und/oder Modellbenutzern relevant scheinen.
3. **Pragmatisches Merkmal:** Modelle sind ihren Originalen nicht per se eindeutig zugeordnet. Sie erfüllen ihre Ersetzungsfunktion a) für *bestimmte* – erkennende und/oder handelnde, modellbenutzende – *Subjekte*, b) innerhalb *bestimmter Zeitintervalle* und c) unter Einschränkung auf *bestimmte gedankliche oder tatsächliche Operationen*

(Stachowiak 1973, 131–33)

- **Mapping aspect** (Abbildungsmerkmal): Models are always models *of something*, viz., mappings from, representations of natural or artificial originals, which can themselves be models.
- **Reductive aspect** (Verkürzungsmerkmal): Models generally *do not capture all attributes* of the original they represent, but only those that seem relevant to the model creators and/or model users.
- **Pragmatic aspect** (Pragmatisches Merkmal): Models are *not per se uniquely assigned* to their originals. They fulfill their replacement function
  1. for particular—cognitive and/or acting—subjects that use the model,
  2. within particular time intervals, and
  3. restricted to particular mental or actual operations.

Note also that while a model does not have all properties of its corresponding original, it has *abundant attributes* (Stachowiak 1973, 155), i.e., attributes that are not present in the original.

Statistics provide us with means to formally describe and analyze a specific subclass of models that are able to represent originals that have particular properties. However, the phenomena studied by the humanities generally do not have these properties, and we thus lack adequate formal methods to describe them.

6 Formalization

Let me say a few more words about formalization. As I have mentioned before, when we consider the epistemology of the digital humanities, we are especially interested in a specific type of formal models, viz. computational models.

Concerning formalization in the humanities, we build in particular on the work of Gilles-Gaston Granger. As for the meaning of the word *formal*, he noted that to formalize is to reduce the language of a theory to primitive expressions and explicit rules of construction.

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> Formaliser, c'est réduire le langage d'une théorie à des expressions primitives et à des règles explicites de construction.
> —Granger ([1960] 1967, 160)

This makes some humanities scholars feel uneasy, because they associate formalization and computation with mathematization, and as Granger ([1960] 1967, 110) has observed, it is still true that "for many who work in the humanities, mathematization is equivalent to the introduction of quantity, indeed of number." This immediately raises the question: but isn't all formal modeling quantitative? Don't the digital humanities mean above all the treatment of large quantities of data, *big data*, and

therefore the use of statistical methods, resulting in the "reduction of phenomena to calculations" (Granger [1960] 1967, 19)?

I don't think so, and also Granger stressed that formalization in the humanities consists in the transition from the unstructured to the structural, rather than in quantification (Granger [1960] 1967, 113).

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The Challenge of Digital Humanities

> Il semble donc qu'une élaboration scientifique des notions qualitatives consiste dans le passage de l'astructuré au structural, bien plutôt que dans une quantification.
> —Granger ([1960] 1967, 113)

As Granger ([1960] 1967, 19) points out, there is a common confusion between formal thought and the work of mathematicians. While it is quite true in a sense that all effective scientific formalism tends towards a mathematical form, this still does not mean that the formalism is unavoidably reduced to the *usual* and *current* instruments of mathematics. And he asks: is it not just this supposed reduction that the opponents of all formalization in the humanities are criticizing? It will be one of the aspects of our task, Granger says, to show formal thought at work in the humanities, not only as a reduction of phenomena to calculations, but also as the invention of new structures, even, indeed, of an original mathematics.

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> À cet égard, il convient de dissiper un préjugé qui nuit souvent à l'appréciation qu'on peut faire de la valeur du formalisme dans les sciences. C'est la confusion que l'on favorise d'ordinaire entre la pensée formelle et l'œuvre des mathématiciens. S'il est bien vrai en un sens que tout formalisme scientifique efficace tend vers un statut mathématique, ce n'est pas pour autant qu'il se réduise infailliblement aux instruments *usuels* et *actuels* des géomètres. N'est-ce pas cependant à cette réduction supposée que s'attaquent essentiellement les adversaires de toute formalisation dans les sciences de l'homme ? Ce sera l'un des aspects de notre tâche que de montrer la pensée formelle à l'œuvre dans les sciences humaines, non pas seulement comme réduction des phénomènes aux calculs, mais aussi comme invention de structures nouvelles, voire même d'une mathématique originale.
> —Granger ([1960] 1967, 19)

I believe that this could really be considered the main challenge of digital humanities.

It also highlights a limitation of current DH research, which is overwhelmingly quantitative, whereas humanities research questions are ultimately qualitative, *why?* questions. The "original mathematics" that Granger envisioned have thus to be understood as "mathematics without numbers" (Kemeny 1959), a general "science of patterns" (Resnik 1999).

A further methodological inspiration comes from Jean-Claude Gardin, whose logicist approach (Gardin 1979) aims to formalize *scholarly constructs*, i.e., the scientific discourse (Gardin 1991), and thus scholars' previously implicit models.

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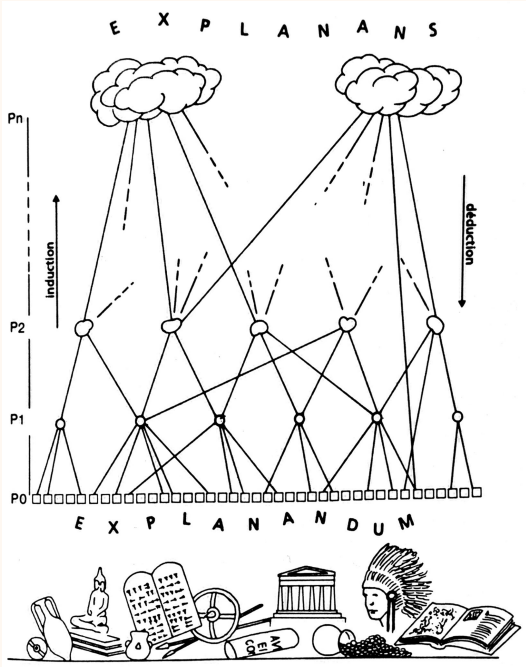
La schématisation des constructions savantes, relatives à toute espèce d'objets (en bas) : chaînes de propositions P_1, P_2... reliant la base de données P_o aux hypothèses P_n.
—Gardin (1991, 65)



Very briefly, Gardin's basic premise is that an argument in the humanities can be reformulated as a chain of propositions P_0, P_1, P_2, ..., P_n, where P_0 designates the given "facts" or database, P_n the hypotheses or conclusions based on them, and P_1, ..., P_{n-1} the inferences that establish P_n on the basis of P_0.

Depending on the chosen mode of argumentation, the chain of propositions can be followed in the empirical-inductive direction, i.e., $P_0 \dashrightarrow P_i \rightarrow P_{i+1} \dashrightarrow P_n$, or in the hypothetical-deductive direction, i.e., $P_n \dashrightarrow P_i \rightarrow P_{i-1} \dashrightarrow P_0$.

As Gardin (1990, 26) makes clear, "the logicist perspective should not be regarded as another manifestation of scientism or neopositivism in the humanities." He stresses that the goal is strictly epistemological—Gardin speaks of "practical epistemology"; in fact, he notes, "the logicist program is an attempt to demonstrate its *limitations* in the humanities" (emph. orig.) in order to determine which parts of the reasoning can, and which cannot be formalized.

7 Conclusion

So far, digital humanities has largely contented itself with borrowing methods from other fields and has developed little methodology of its own. The almost exclusive focus on methods and tools represents a major obstacle towards the construction of computational models that could help us to obtain new insights into *humanities* research questions rather than just automate primarily quantitative processing—which is, without doubt, useful, but inherently limited, given that the research questions are ultimately qualitative.

I have outlined some epistemological issues in digital humanities and have sketched some potential approaches for addressing them.

Regardless of the application domain, digital humanities research tends to rely heavily on *corpora*, i.e., curated collections of texts, images, music, or other types of data. However, the epistemological implications have so far been largely ignored. I have proposed to consider corpora as *phenomenotechnical devices* (Bachelard [1934] 1968), like scientific instruments: corpora are, on the one hand, models of the phenomenon under study; on the other hand, the phenomenon is *constructed* through the corpus. We therefore need to study **corpora as models** to answer questions such as: How do corpora model and produce phenomena? What are commonalities and differences between different types

of corpora? How can corpora-as-models be formally described in order to take their properties into account for research that makes use of them?

This is all work in progress, but I believe that the combination of Bachelard's *phénoménotechnique*, Granger's theoretical work on formalization in the humanities, and the logicist approach of Gardin constitute a solid basis for studying the epistemological questions I have outlined, as well as others that I could not discuss today.

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